

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: EET205

Course Name: ANALOG ELECTRONICS

Scheme of Valuation/Answer Key

PART A

Answer all questions. Each question carries 3 marks

1. Biasing is the process of setting the dc operating voltage (or) current condition to the correct level so that any ac input signal can be amplified correctly by the transistor.

NEED FOR BIASING

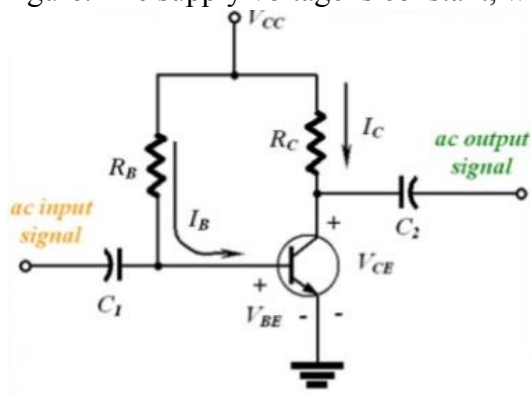
It should ensure proper zero signal collector current.

It should ensure proper V_{BE} at any instant

I_E ; 0.7V for Si transistor and 0.3V for Ge transistor

It should ensure that V_{CE} does not fall below 0.5 V for Ge transistors and 1 V for silicon transistors at any instant.

2. Fixed bias is the simplest transistor DC bias configuration as shown in the figure. The supply voltage is constant, which is chosen in advance



Applying KVL to the base-emitter loop, we get

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B R_B = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \rightarrow \textcircled{1}$$

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

Disadvantages :-

- 1) Flow of current in collector circuit produces heat at the collector junction. This increases the temperature of the junction i.e.

I_{CBO} doubles for every 10°C rise in temperature.

$$I_C = \beta I_B + I_{CEO}$$

- 2) Since $I_C = \beta I_B$ & I_B is already fixed; I_C depends on β which changes from transistor to transistor & shifts the operating point.

Thus Stabilization of operating point is very poor in the fixed bias circuit.

3. At low f

The reactance of input capacitor C_{in} and the coupling capacitor C_C are so high that only small part of the input signal is allowed.

The reactance of the emitter bypass capacitor C_E is also very high during low frequencies. Hence it cannot shunt the emitter resistance effectively.

With all these factors, the voltage gain falls off at low frequencies

At High f

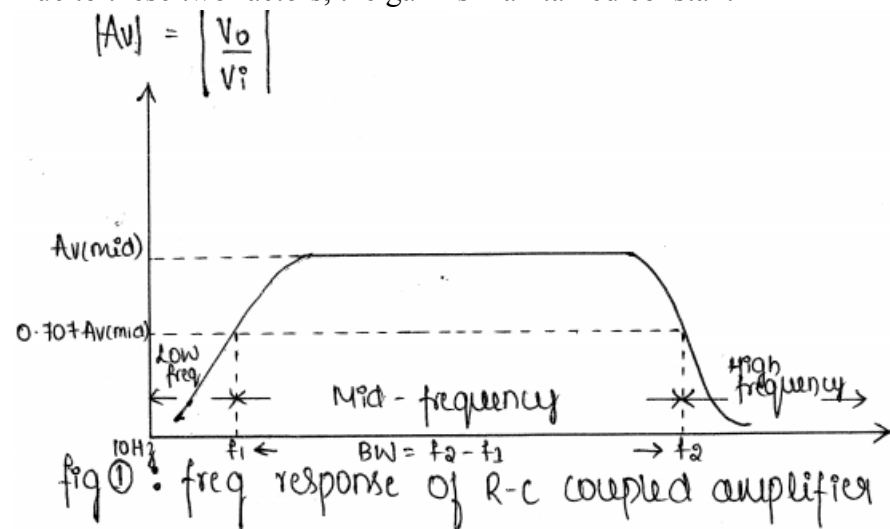
- As a result of this, the loading effect of the next stage increases, which reduces the voltage gain.
- Along with this, as the capacitive reactance of BE junction decreases, it increases the base current of the transistor due to which the current gain ($\beta = I_C/I_B$) reduces.

Hence the voltage gain rolls off at high frequencies

At Mid f

- The voltage gain of the capacitors is maintained constant in this range of frequencies, as shown in figure. If the frequency increases, the reactance of the capacitor C_C decreases which tends to increase the gain.
- But this lower capacitance reactive increases the loading effect of the next stage by which there is a reduction in gain.

Due to these two factors, the gain is maintained constant



4.

A JFET is a three terminal semiconductor device in which current conduction is by one type of carrier i.e. electrons or holes.

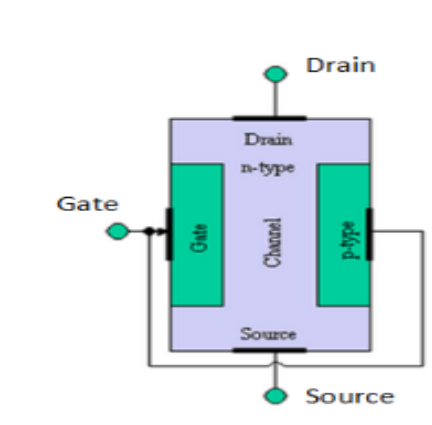
The current conduction is controlled by means of an electric field between the gate and the conducting channel of the device.

The JFET has high input impedance and low noise level.

A JFET consists of a p-type or n-type silicon bar containing two pn junctions at the sides as shown in fig.

The bar forms the conducting channel for the charge carriers.

If the bar is of p-type, it is called p-channel JFET and if the bar is of n-type, it is called n-channel JFET



5. An oscillator is basically an amplifier which does not have any ac input signal but it operate on the principle of positive feedback to generate ac signal at its output. Positive feedback does not always guarantees oscillations. An amplifier will work as an oscillator if and only if it satisfies a set of conditions called Barkhausen criterion. According to **Barkhausen criterion** an amplifier will work as an oscillator if and only if its total **phase difference** should be **zero or multiple of 2π** and **$|A\beta| = 1$** i.e magnitude of feedback factor should be unity.

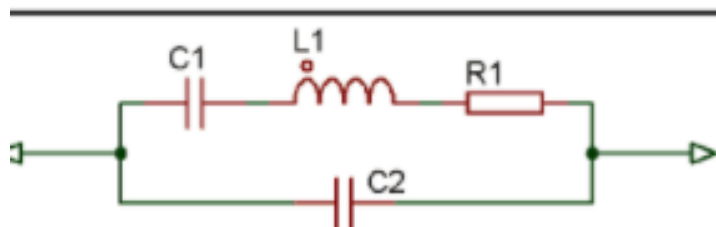
The gain of the positive feedback amplifier is given by

$$A_f = \frac{A}{(1 - A\beta)}$$

6.



Symbol of Quartz Crystal



7.

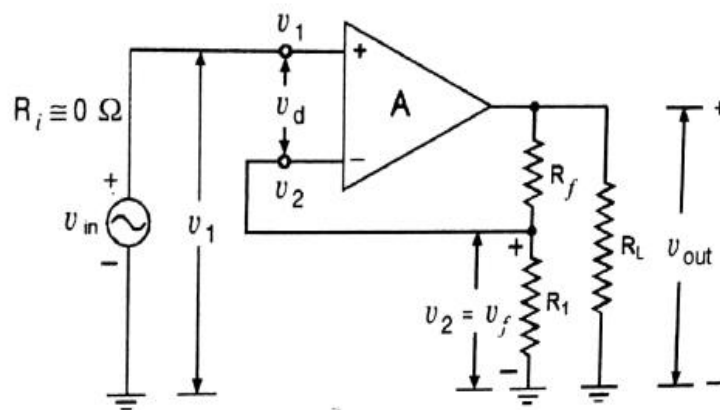
1. Infinite voltage gain.
2. Infinite input impedance.
3. Zero output impedance.
4. Zero offset voltage.
 - The presence of small output voltage though $V_1=V_2=0$.
5. Infinite bandwidth.
6. Infinite CMRR.
7. Infinite slew rate.
8. feedback voltage V_f (ie voltage at V_2) is almost equal to the input voltage V_{in} (Voltage at V_1), since $V_1 - V_2$ approximately equal to zero for op – amp, then $V_1 - V_2 = V_{in} - V_f$

$$V_{in} = V_2 = V_f = \frac{V_o R_1}{R_1 + R_f}$$

$$\frac{V_{in}}{V_o} = \frac{R_1}{R_1 + R_f}$$

$$A_{CL} = Af = \frac{V_o}{V_{in}} = \frac{R_1 + R_f}{R_1}$$

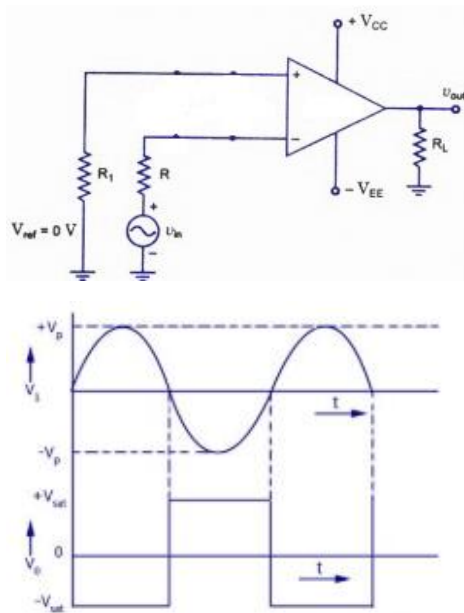
$$Af = 1 + \frac{R_f}{R_1}$$



$$\frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_1}$$

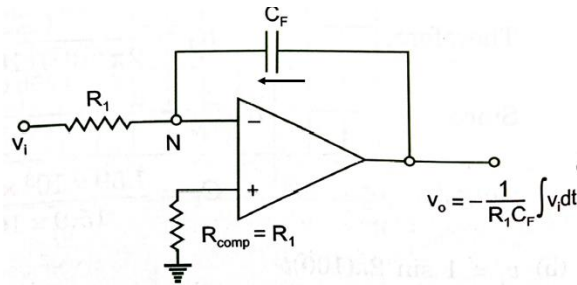
$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

9.



- Zero crossing detector is a comparator circuit in which the reference voltage is zero.
- The zero crossing detector produces pure square wave signal for a input sine wave. So it is also called as Sine to square wave generator.
- The inverting zero crossing detector circuit is shown in figure.
- Input signal is given to the inverting input terminal and the reference voltage is set to zero.
- The input is applied to the inverting input terminal and the non inverting input is connected to ground
- When the input is positive the output is $-V_{sat}$
- When the input is negative the output is $+V_{sat}$.

10.

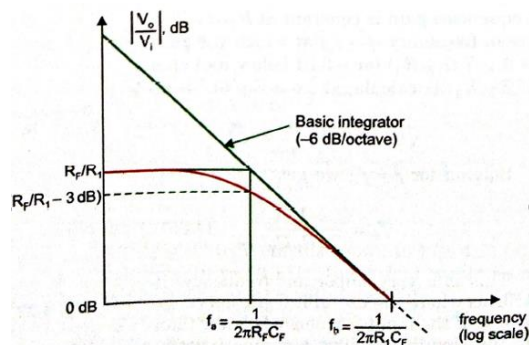


$$\frac{v_i}{R_1} + C_F \frac{dv_o}{dt} = 0$$

or,
$$\frac{dv_o}{dt} = -\frac{1}{R_1 C_F} v_i$$

Integrating both sides, we get

$$\int_0^t dv_o = -\frac{1}{R_1 C_F} \int_0^t v_i dt$$

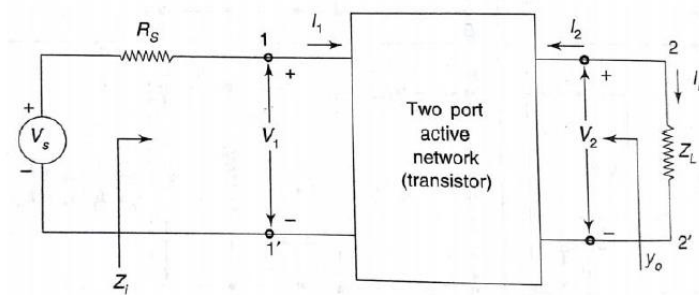


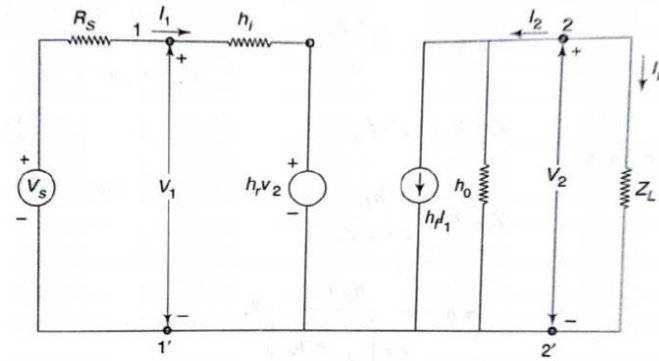
PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11a.





$$A_I = \frac{I_L}{I_1} = \frac{-I_2}{I_1} \dots\dots(1)$$

$$-I_2 = A_I I_1 \dots\dots(2)$$

- From the above Equation

$$I_2 = h_f I_1 + h_o V_2 \dots\dots(3)$$

$$V_2 = I_L R_L = -I_2 R_L \dots\dots(4)$$

- Substitute Equation (4) in (3)

$$I_2 = h_f I_1 + h_o (-I_2 R_L)$$

$$I_2 + I_2 R_L h_o = h_f I_1$$

$$I_2 [1 + R_L h_o] = h_f I_1$$

$$I_1 = \frac{I_2 (1 + R_L h_o)}{h_f}$$

$$\frac{I_2}{I_1} = \frac{h_f}{1 + R_L h_o}$$

$$A_I = \frac{-I_2}{I_1} = \frac{-h_f}{1 + R_L h_o}$$

Input Impedance (z_i)(or) (R_i)

- In the circuit R_s is the signal source resistance

$$Z_i = R_i = \frac{V_i}{I_i} = \frac{V_1}{I_1}$$

- From the above equation

$$V_1 = h_i I_1 + h_r V_2$$

$$Z_i = R_i = \frac{V_1}{I_1} = \frac{h_i I_1 + h_r V_2}{I_1} = h_i + h_r \frac{V_2}{I_1}$$

Voltage Gain (or) Voltage amplification factor

$$V_2 = -I_2 R_L$$

$$A_I = \frac{-I_2}{I_1}$$

$$-I_2 = A_I I_1$$

$$V_2 = A_I I_1 R_L$$

$$R_i = h_i + h_r \frac{A_I I_1}{I_1} R_L$$

$$R_i = h_i + h_r A_I R_L$$

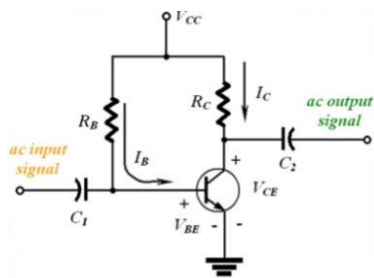
- The ratio of output voltage V_2 to input voltage V_1 gives the voltage gain of the transistor.

$$A_V = \frac{V_2}{V_1}$$

$$V_2 = -I_2 R_L = A_I I_1 R_L$$

$$A_V = \frac{A_I I_1 R_L}{V_1} = \frac{A_I R_L}{R_i}$$

11b.



Applying KVL to the base-emitter loop, we get

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B R_B = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \rightarrow \textcircled{1}$$

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

Given $\beta=80$, $R_B=100\text{k}\Omega$ and $R_C=10\text{k}\Omega$ and $V_{CC}=15\text{V}$

$I_C = 1.27\text{mA}$, $V_{CE} = 2.3\text{V}$ - 3 Marks

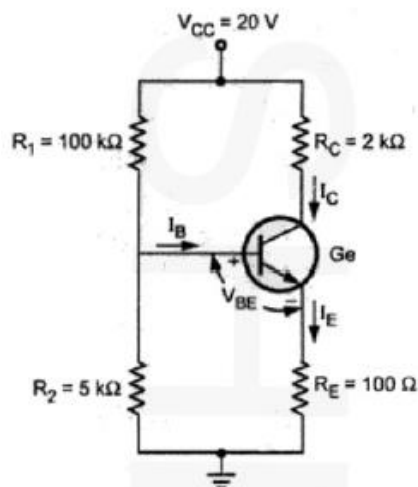
12a.

$$V_{TH} = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (1 + \beta) R_E}$$

$$V_{CE} = V_{CC} - I_C [R_C + R_E]$$

$R_1=100\text{k}\Omega$, $R_2=5\text{k}\Omega$ and $R_E=100\Omega$. Calculate I_B , V_{CE} , I_C and stability factor, S .



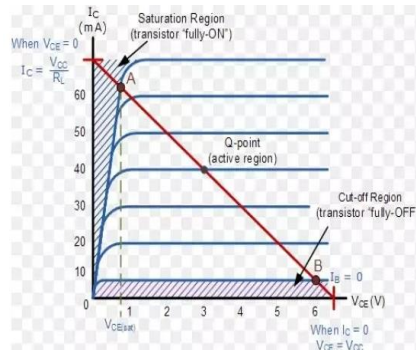
$I_B=76.29\mu\text{A}$ (2 marks)

$I_C=3.814\text{ mA}$ (2 mark for final answer)

$V_{CE}=11.983\text{V}$ (3)

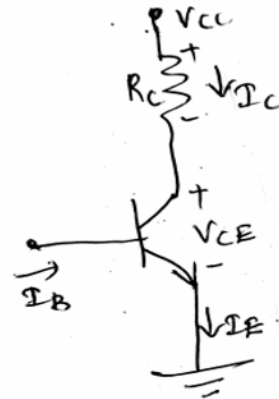
Stability factor, $S=25.143$ (3)

12b.



$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CC} = I_C R_C + V_{CE}$$



- When collector Emitter voltage $V_{CE} = 0$, then collector current is maximum ($I_{C \text{ sat}}$) and is equal to $\frac{V_{CC}}{R_C}$

- i.e.,

$$V_{CC} = I_C R_C + V_{CE}$$

$$V_{CE} = 0$$

$$V_{CC} = I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C}$$

- So the point is $\left(0, \frac{V_{CC}}{R_C}\right)$

- (i) When the collector current $I_C = 0$, then Collector Emitter voltage is maximum and is equal to V_{CC} . i.e.,

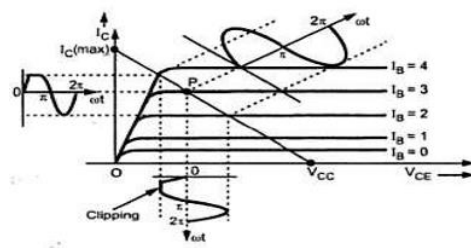
$$V_{CC} = I_C R_C + V_{CE}$$

$$I_C = 0$$

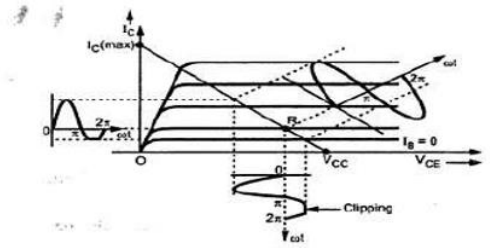
$$V_{CC} = V_{CE}$$

- So the point is

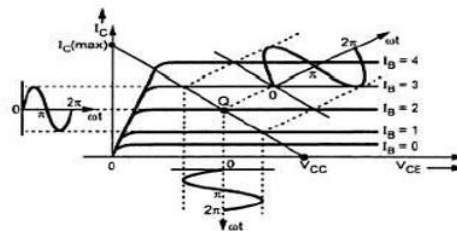
$$(V_{CC}, 0)$$



Operating point near saturation region gives clipping at the positive peaks

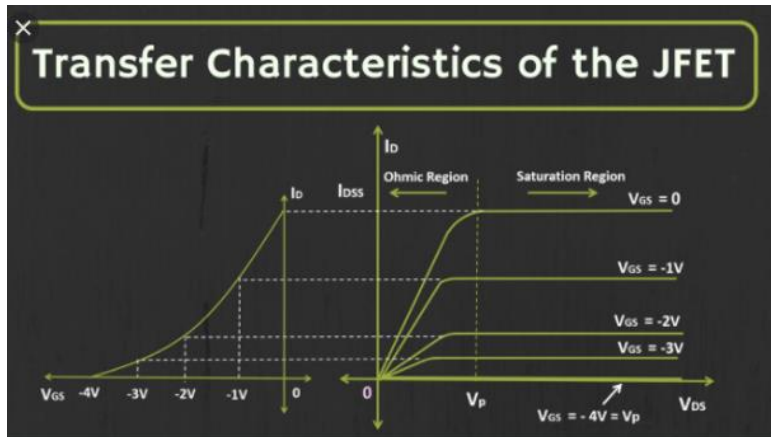


Operating point near cut-off region gives clipping at the negative peaks



Operating point at the centre of active region is most suitable

Module 2



13b.

Amplification factor(μ)

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}} / I_D \text{ Constant}$$

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

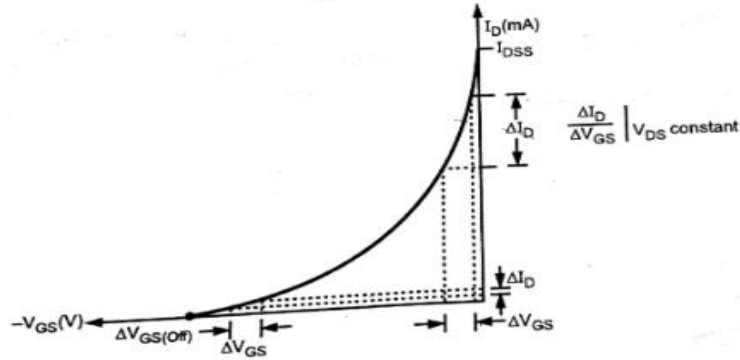
$$\mu = \frac{\Delta V_{DS}}{\Delta I_D} * \frac{\Delta I_D}{\Delta V_{GS}}$$

$$\mu = r_d * g_m$$

Since the parameter μ is the ratio of two similar quantities viz. ratio of two voltages. μ is unit less.

Input resistance (R_i): JFET operates with its gate source junction is reverse biased. That is gate reverse current of JFET is very small. The gate source junction is used here as the input. So input resistance is very high. This input Resistance is an advantage of JFET

$$R_i = V_{GS} / I_{GSS}$$



The transconductance g_m , is the change in the drain current for given change in gate to source voltage with the drain to source voltage constant

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \bigg|_{V_{DS} \text{ constant}}$$

The transconductance g_m is also called mutual conductance

$$g_m = g_{mo} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)$$

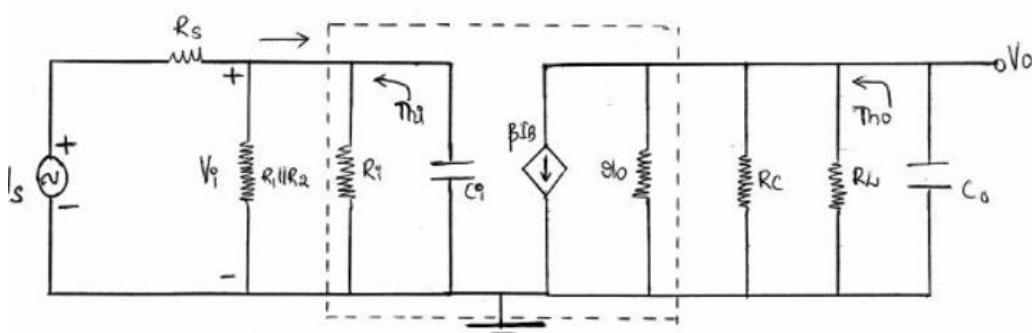
Where g_{mo} is the value of g_m for $V_{GS} = 0$ and is given by

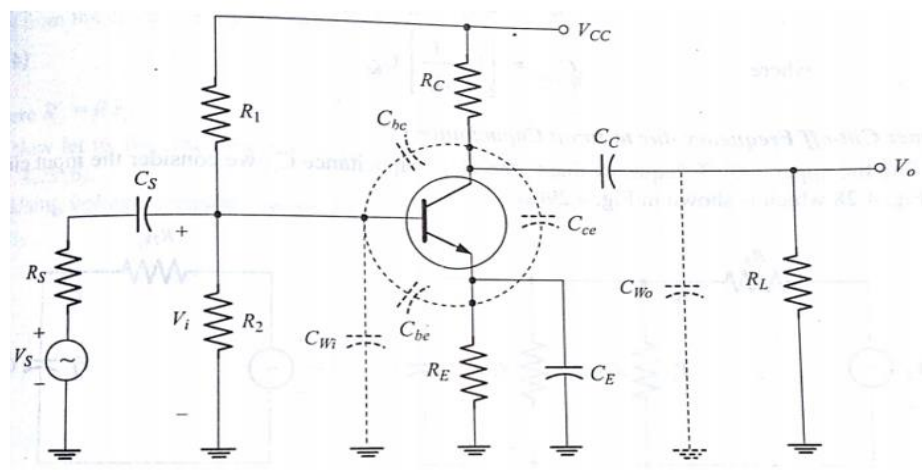
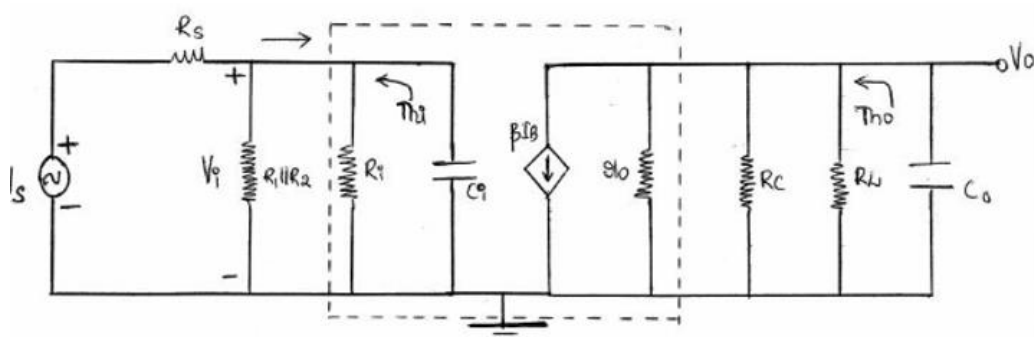
$$g_{mo} = \frac{-2I_{DSS}}{V_P}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

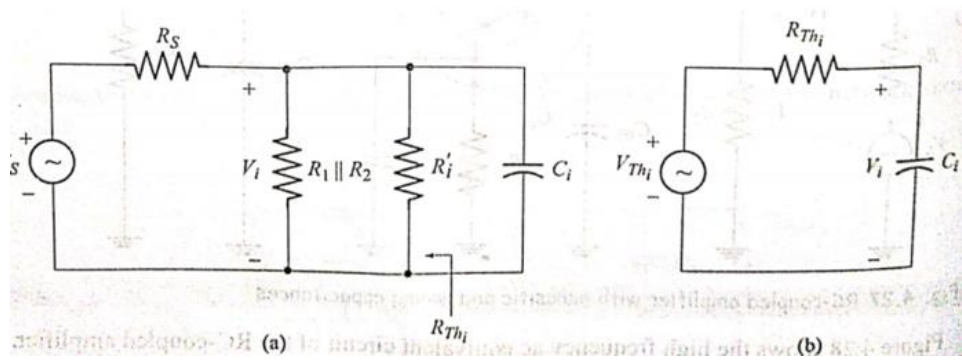
$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{-2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right)$$

14 a.





- Consider input ckt ,



→ Express amplifier V_i as function of frequency from ckt 4.29(b).

→ By v/g division Rule.

$$|V_i| = \frac{|V_{Thi}| X_{Ci}}{\sqrt{(R_{Thi})^2 + (X_{Ci})^2}}$$

$$|V_i| = \frac{|V_{Thi}|}{\sqrt{1 + \left(\frac{R_{Thi}}{X_{Ci}}\right)^2}}$$

where $X_{Ci} = \frac{1}{2\pi f C_i}$.

- For input network, the 3dB frequency is defined by

•

$$|V_i| = \frac{|V_i|_{mid}}{\sqrt{2}} = \frac{|V_{Thi}|}{\sqrt{2}}$$

-where input voltage in the mid frequency range $|V_i|_{mid} = |V_{Thi}|$

- From the equation of $|V_i|$ we find that this condition occurs when

$$R_{Thi} = X_{Ci}$$

$$R_{Thi} = \frac{1}{2\pi f C_i}$$

$$f = \frac{1}{2\pi R_{Thi} C_i}$$

- Above equation gives upper 3dB cut off frequency due to C_i

$$f_{Hi} = \frac{1}{2\pi R_{Thi} C_i}$$

- Consider output circuit

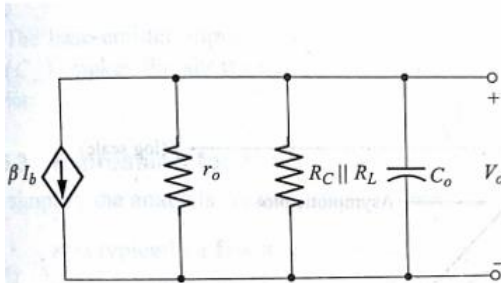


Fig. 4.32 Output ac equivalent circuit

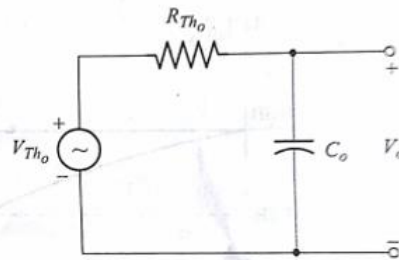


Fig. 4.33 Simplified output ac equivalent circuit

The current source βI_b along with the resistors r_o and $R_C \parallel R_L$ can be converted into its equivalent voltage source as shown in Fig. 4.33.

$$R_{Th_o} = r_o \parallel R_C \parallel R_L$$

$$V_{Th_o} = [-\beta I_b] [r_o \parallel R_C \parallel R_L]$$

The output voltage as a function of frequency is

$$|V_o| = \frac{|V_{Th_o}|}{\sqrt{1 + \left[\frac{R_{Th_o}}{X_{C_o}} \right]^2}}$$

where

$$X_{C_o} = \frac{1}{2\pi f C_o}$$

The output voltage in the mid band is

$$|V_o|_{\text{mid}} \approx |V_{Th_o}|$$

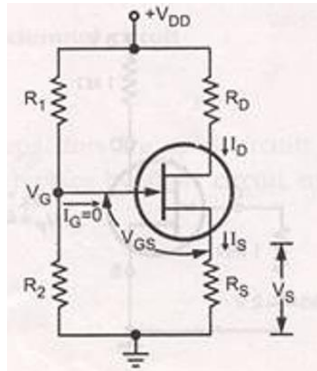
The upper 3dB cutoff frequency due to C_o is

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$

Combined effect of C_i and C_o on high frequency response

- The Input capacitance C_i defines the upper cut off frequency f_{H_i}
- The Output capacitance C_o defines another upper cut off frequency f_{H_o}
- The lowest of these two frequencies will be taken as overall upper – cut off frequency.

14b. For a JFET connected in voltage divider biasing circuit, calculate I_D , V_{DS} and V_{GS} with $V_{DD}=24V$, $R_1= 910k\Omega$, $R_2= 110k\Omega$, $R_D= 22k\Omega$, $R_S= 1.1k\Omega$, $I_{DSS}= 10mA$ and pinch-off voltage of the JFET is $3.5V$



$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$V_G - V_{GS} - V_S = 0$$

$$V_G - V_{GS} - I_S R_S = 0$$

$$V_{GS} = V_G - I_S R_S$$

$$I_D = I_S$$

$$V_{GS} = V_G - I_D R_S$$

$$V_{GS} = V_G - I_D R_S$$

$$I_D = 0$$

$$V_{GS} = V_G$$

$$V_{GS} = 0$$

$$0 = V_G - I_D R_S$$

$$I_D = \frac{V_G}{R_S}$$

$$V_G = 2.588V \text{ (1 Mark)}$$

$$I_D = 8.471mA, 3.616mA \text{ (4 Marks)}$$

$$V_{gs} = -6.7301V \text{ (for } 8.471mA), -1.3896V \text{ (for } 3.616mA) \text{ (2 Marks)}$$

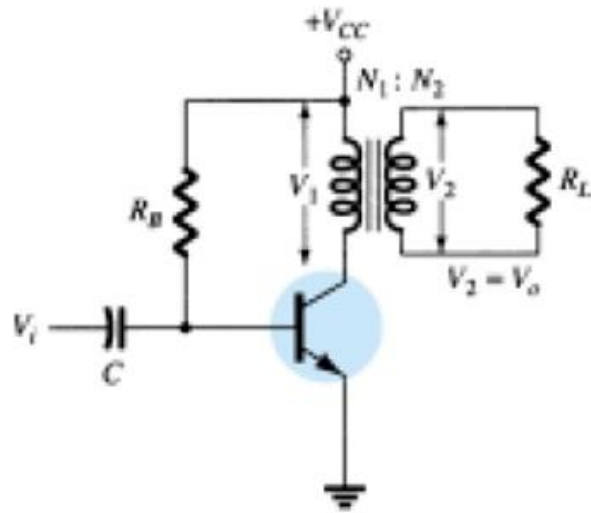
$$V_{DS} = 12.067V \text{ (1 Mark)}$$

Module 3

15a.

- Effectiveness of power amplifier is measured by its capability of converting dc to ac power.

$$\eta = \frac{\text{average a. c power output}}{\text{average d. c power input to transistor}}$$



(a)

$$\frac{I_2}{I_1} = \frac{N_1}{N_2}$$

$$\frac{R_L}{R_1} = \frac{R_2}{R_1} = \frac{\frac{V_2}{I_2}}{\frac{V_1}{I_1}} = \frac{V_2 I_1}{V_1 I_2} = \left(\frac{N_2}{N_1} \right)^2$$

$$\frac{R_L}{R_1} = \frac{R_2}{R_1} = \left(\frac{N_1}{N_2} \right)^2 = a^2$$

$$\frac{R_1}{R_2} = a^2$$

(or)

$$R_1 = a^2 R_L$$

$$\frac{R_L}{R_1} = a^2$$

(or)

$$R_L = a^2 R_1$$

$$V_{CE(p-p)} = 2V_{CE(p)} = 2V_{CC}$$

$$I_{C(p-p)} = 2I_C = \frac{V_{CE(p-p)}}{R_L}$$

$$P_{O(ac)} = \frac{V_{CE(p-p)} I_{C(p-p)}}{8}$$

$$P_{O(ac)} = \frac{2V_{CC} 2I_C}{8} = \frac{1}{2} * V_{CC} I_C$$

$$\% \eta = \frac{P_{O(ac)}}{P_{i(dc)}} = \frac{\frac{1}{2} * V_{CC} I_C}{V_{CC} I_C} * 100\% = 50\%$$

15b.

- They have excellent audio fidelity over a wide range of frequency.
- Widely used as Voltage amplifiers
- Due to poor impedance matching, RC coupling is rarely used in the final stages.

16a.

- When an AC input signal is applied to the base of first transistor, it gets amplified and appears at the collector resistor R_C which is then passed through the coupling capacitor C_C to the next stage.
- This becomes the input of the next stage, whose amplified output again appears across its collector load. Thus the signal is amplified in stage by stage action.
- Overall gain is less than the product of gains of individual stages due to loading effect (load resistance of first stages reduced due to shunting effect of second stage input R_i)

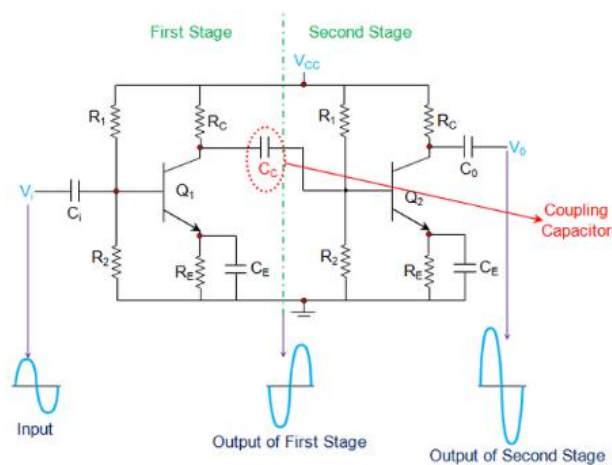
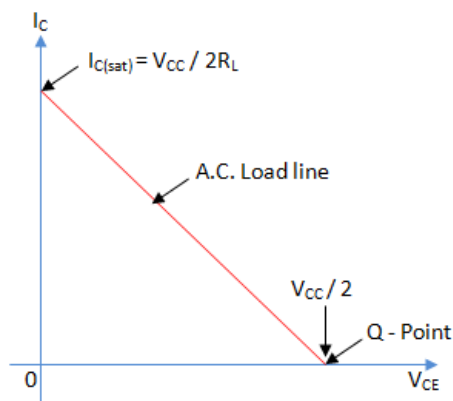


Figure 1 Two-Stage RC Coupled Amplifier

- As we consider a two stage amplifier here, the output phase is same as input. Because the phase reversal is done two times by the two stage CE configured amplifier circuit

16b.



- It is clear that a.c. output voltage has a peak value of V_{CE} and a.c. output current has a peak value of $I_{C(sat)}$

$$\therefore \text{Peak a. c. output voltage} = V_{CE}$$

$$\text{Peak a. c. output current} = I_{C(sat)} = \frac{V_{CE}}{R_L}$$

- Maximum average a.c. output power $P_{o(max)}$ is
- $P_{o(max)}$ = Product of r.m.s. values of a.c. output voltage and a.c. output current

$$\begin{aligned} &= \frac{V_{CE}}{\sqrt{2}} \times \frac{I_{C(sat)}}{\sqrt{2}} = \frac{V_{CE} I_{C(sat)}}{2} \\ &= \frac{V_{CC}}{2} \times \frac{I_{C(sat)}}{2} = \frac{V_{CC} I_{C(sat)}}{4} \end{aligned}$$

- . The input d.c. power from the supply V_{CC} is

$$\text{Where } I_{dc} \text{ is the average current drawn from the supply } V_{CC} \quad P_{dc} = V_{CC} I_{dc}$$

$$\therefore I_{dc} = \frac{I_{C(sat)}}{\pi}$$

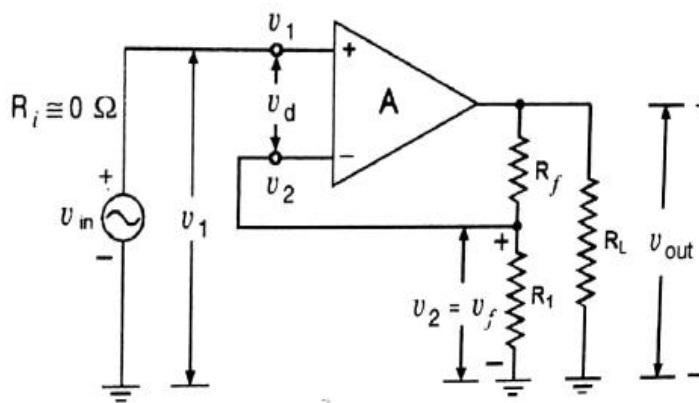
$$\therefore P_{dc} = \frac{V_{CC} I_{C(sat)}}{\pi}$$

$$\therefore \text{Max. collector } \eta = \frac{P_{o(max)}}{P_{dc}} = \frac{0.25V_{CC}I_{C(sat)}}{(V_{CC}I_{C(sat)})/\pi} \times 100 = 0.25\pi \times 100 = 78.5\%$$

- Thus, the maximum collector efficiency of class B power amplifier is 78.5% .

Module 4

17a.



$$V_{in} = V_2 = V_f = \frac{V_o R_1}{R_1 + R_f}$$

$$\frac{V_{in}}{V_O} = \frac{R_1}{R_1 + R_f}$$

$$A_{CL} = Af = \frac{V_o}{V_{in}} = \frac{R_1 + R_f}{R_1}$$

$$Af = 1 + \frac{R_f}{R_1}$$

$$\frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_1}$$

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

17b.

Advantages of Negative Voltage Feedback

- **Gain stability.**

- An important advantage of negative voltage feedback is that the resultant gain of the amplifier can be made independent of transistor parameters or the supply voltage variation

$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

- For negative voltage feedback in an amplifier to be effective, the designer deliberately makes the product $A_v m_v$ much greater than unity. Therefore, in the above relation, 1 can be neglected as compared to $A_v m_v$ and the expression becomes :

$$A_{vf} = \frac{A_v}{A_v m_v} = \frac{1}{m_v}$$

- It may be seen that the gain now depends only upon feedback fraction m_v i.e., on the characteristics of feedback circuit.

- **Reduces non-linear distortion.**

- A large signal stage has non-linear distortion because its voltage gain changes at various points in the cycle. The negative voltage feedback reduces the nonlinear distortion in large signal amplifiers.

- **Improves frequency response.**

- As feedback is usually obtained through a resistive network, therefore, voltage gain of the amplifier is independent of signal frequency. The result is that voltage gain of the amplifier will be substantially constant over a wide range of signal frequency. The negative voltage feedback, therefore, improves the frequency response of the amplifier.

- **Increases circuit stability.**

- **Increases input impedance and decreases output impedance.**

18.a

A differential amplifier has a gain of 100. A common input of 5mV is applied to both terminals, which result in an output voltage of 18mV. Determine i) common mode gain ii) CMRR. If the input signals are changed to 50mV and 100mV with 1mV of noise on each input. Find iii) the output signal iv) the noise on the output

$$CMRR = \frac{A_{DM}}{A_{CM}}$$

$$A_{DM} = \frac{V_o}{V_{in}}$$

$$A_{CM} = \frac{R_C}{2 R_E}$$

common mode gain = 3.6

CMRR = 28.87dB

the output signal = 5.0036V

18b.

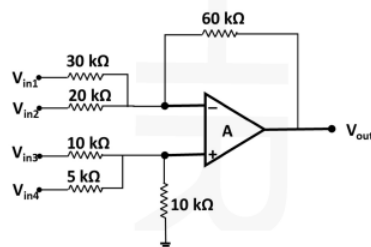


Fig.2

$$R_1 = R_2$$

$$V_o = -\frac{R_f}{R_1} V_1 + \left(1 + \frac{R_f}{R_1}\right) \left(\frac{R_f}{R_1 + R_f}\right) V_2$$

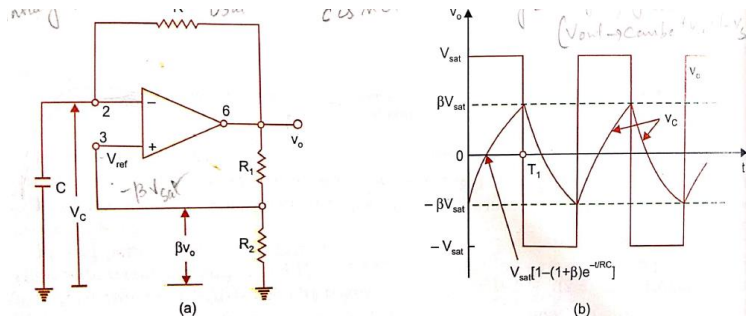
$$V_o = -\frac{R_f}{R_1} V_1 + \left(\frac{R_1 + R_f}{R_1}\right) \left(\frac{R_f}{R_1 + R_f}\right) V_2$$

$$V_o = \frac{R_f}{R_1} (V_2 - V_1)$$

$$V_o = -(2V_{in1} + 3V_{in2}) + (1.5V_{in3} + 3V_{in4})$$

Module 5

19a.



$$V_o(t) = V_f + (V_i - V_f) e^{-t/\tau} \rightarrow (1)$$

$$\text{let } V_f = +V_{sat}$$

$$V_i = -\beta V_{sat}$$

$\therefore$ (1) becomes.

$$V_o(t) = V_{sat} + (-\beta V_{sat} - V_{sat}) e^{-t/\tau}$$

$$V_o(t) = V_{sat} - V_{sat}(1+\beta) e^{-t/\tau}$$

$$V_o(t) = V_{sat} (1 - (1+\beta) e^{-t/\tau}) \rightarrow (2)$$

$$\rightarrow \text{At } t = T_1, V_o(t) = \beta V_{sat} \text{ (fig.)}$$

$\therefore$ eq (2).

$$V_o(T_1) = \beta V_{sat} = V_{sat} (1 - (1+\beta) e^{-T_1/\tau})$$

$$\Rightarrow \beta = 1 - (1+\beta) e^{-T_1/\tau}$$

$$e^{-T_1/\tau} = \frac{1-\beta}{1+\beta}$$

$\rightarrow \log e$ on both sides

$$-T_1/\tau = \ln \frac{1-\beta}{1+\beta}$$

Charging Time

$$T_1 = -RC \ln \left(\frac{1-\beta}{1+\beta} \right) = \frac{2RC}{\beta}$$

$$T = 2T_1 = 2RC \ln \left(\frac{1+\beta}{1-\beta} \right)$$

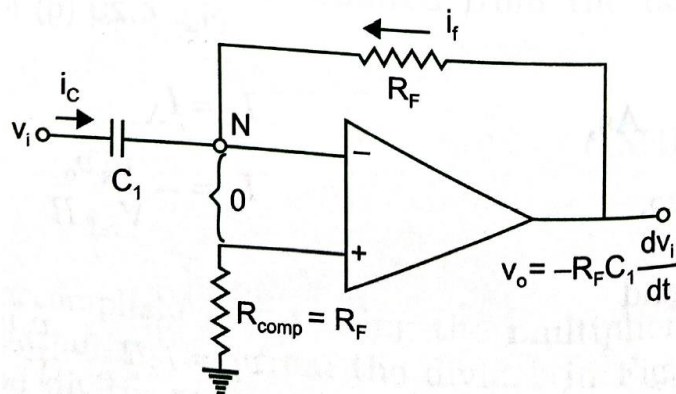
$$T = 2RC \ln \frac{R_1 + 2R_2}{R_1} \text{ Sec.}$$

$$f_o = \frac{1}{T} = \frac{1}{2RC \ln \left(\frac{R_1 + 2R_2}{R_1} \right)}$$

if $R_1 = 1.16 R_2$
we get $f_o = \frac{1}{2RC}$, $f_o \propto \frac{1}{R}$ if C constant

19b.

$$i_C = C_1 \frac{d}{dt} (v_i - v_N) = C_1 \frac{dv_i}{dt}$$



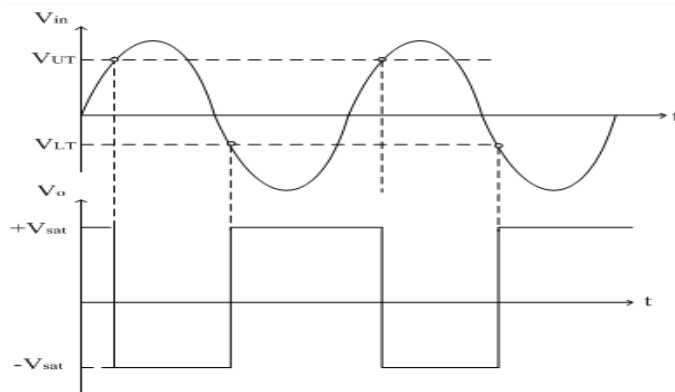
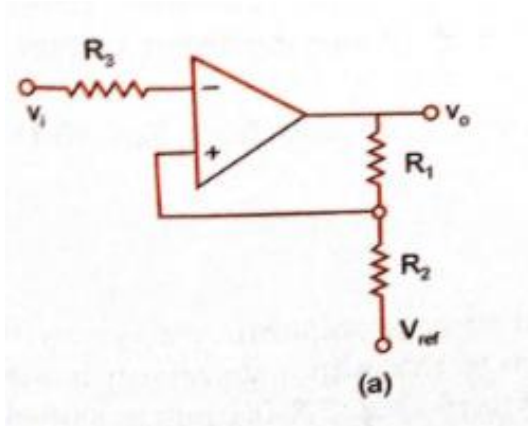
The current i_f through the feedback resistor is v_o/R_f and there is no current into the op-amp. Therefore, the nodal equation at node N is,

$$C_1 \frac{dv_i}{dt} + \frac{v_o}{R_F} = 0$$

from which we have

$$v_o = -R_F C_1 \frac{dv_i}{dt}$$

20a.

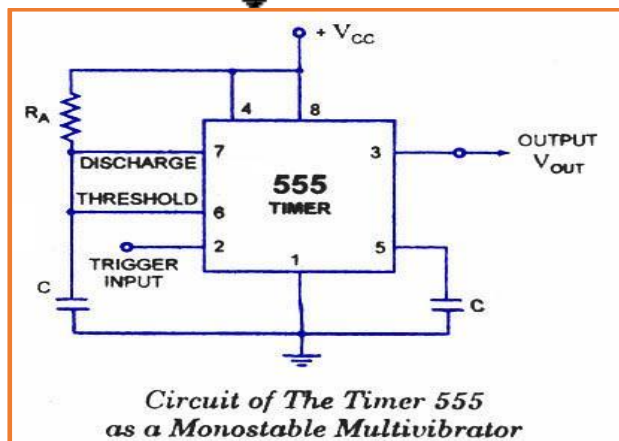
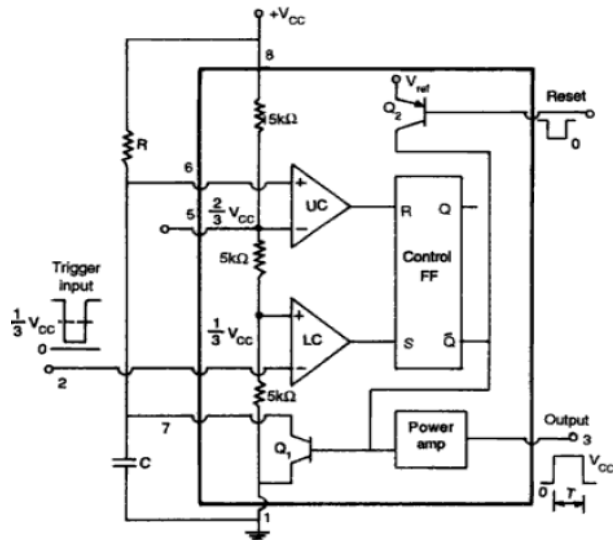


- The input voltage V_i triggers the output V_o every time when it exceeds certain voltage levels. These voltage levels are called upper threshold voltage and lower threshold voltage.
- The hysteresis width is the difference between these two threshold voltages.
- Suppose $V_o = +V_{sat}$ the voltage at the (+)input terminal will be

$$V_{ref} + \frac{R_2}{R_1 + R_2} (V_{sat} - V_{ref}) = V_{UT}$$

- As long as V_i is less than V_{UT} , the output V_o remains constant at $+V_{sat}$. When V_i is just greater than V_{UT} . The output regeneratively switches to $-V_{sat}$ and remains at this level as long as V_i is greater than V_{UT}

20b.



- Voltage across C rises exponentially through R towards V_{cc} with a time constant RC .
- After Time Period T , the capacitor voltage is just greater than $2V_{cc}/3$ and the upper comparator resets the FF, i.e. $R=1$, $S=0$ (note that, after the application of trigger, pin 2 will be at V_{cc}). This makes $Q\bar{=1}$, C discharges rapidly to ground potential.

The voltage across the capacitor as given by

$$v_C = V_{cc}(1 - e^{-t/RC})$$

$$t = T, v_C = \frac{2}{3}V_{cc}$$

$$\frac{2}{3}V_{cc} = V_{cc}(1 - e^{-T/RC})$$

$$T = RC \ln\left(\frac{1}{3}\right) \Rightarrow T = 1.1RC \text{ sec}$$

If -ve going reset pulse terminal (pin 4) is applied, then transistor $Q_2 \rightarrow$ OFF, $Q_1 \rightarrow$ ON & the external timing capacitor C is immediately discharged